Glottal Inverse Filtering Using Probabilistic Weighted Linear Prediction

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***Abstract*—Glottal inverse filtering is a noninvasive method for getting the glottal flow estimate from the speech. In this paper, we propose a method for glottal inverse filtering based on probabilistic weighted linear prediction (PWLP) in which the speech is assumed to be the output of an all-pole filter with glottal flow as an excitation. First, we introduce a probabilistic interpretation of the WLP, and we propose a probabilistic temporal weighting as convolution of a binary vector and a fixed window. We construct the posterior distribution based on the PWLP likelihood and a Gaussian prior on the filter coefficients. The parameters are estimated using the Gibbs sampling. The experiments are performed using the Lijencrants–Fant (LF) model based synthetic data, a physical model based synthetic data of different vowels and real speech data. Results demonstrate that the proposed method outperforms the best of the existing state-of-the-art methods in terms of the normalized amplitude quotient by 0.035 and 0.12 for the LF model and physical model based synthetic data, respectively. The results based on real speech data show that the glottal flow estimated by the proposed method in the closed phase is flatter and has less formant ripple compared to existing state-of-the-art methods. We also show two key features of the proposed method: first, the proposed method does not need prior detection of glottal closure or opening instants. The temporal weights are learnt in a data-driven manner, which is often found to be high near the closed phase of the glottal cycle, second, the Gaussian prior helps in estimating the filter coefficients when the closed phase duration is small.**

***Index Terms*—Glottal inverse filtering, GIF, Gibbs sampling, probabilistic weighted linear prediction.**

I. Introduction

SPEECH production is often modeled by the source filter model. Source filter model assumes that the speech is generated by exciting a filter with an input source signal. The speech sounds are broadly classified into two categories based on the type of excitation [1] : 1) Voiced sound where the excitation is a quasi periodic flow of air generated by the vibration of the vocal folds and 2) Unvoiced sound where the filter is excited by white noise. The filter is often assumed to model the effect of the vocal tract tube and the lip radiation. The glottal velocity signal is useful in several applications such as medicine (e.g., occupational voice [2] or speech pathology [3]), phonetics (e.g., prosody [4]), and neuroscience (e.g., studying brain responses evoked by speech [5]) and the speech synthesis [6]. However, the direct measurement of the glottal flow is very difficult because the vocal folds are located in the larynx. Hence, the glottal flow velocity is estimated from the speech signal. The process of estimating glottal flow velocity from the speech is known as the Glottal inverse filtering (GIF). The GIF methods cancel the effect of vocal tract and the lip radiation to get the glottal flow estimate [7].

The main part of the GIF is to accurately estimate the effect of vocal tract filter and the lip radiation. Various properties of the glottal flow signal are exploited to get the accurate estimate of the filter. The sample glottal flow and its first derivative are shown in Fig. 1 and three stages over the glottal cycle are marked. The first stage is called the opening stage, which is the glottal flow from the instant at which glottis opens to the instant at which the glottal velocity reaches the peak. The second stage is called the closing stage, which begins at the end of the opening phase and continues up to the instant where the glottal flow becomes zero. In the final stage the glottis stays closed, which is referred to as the closed phase. In the closed phase the glottal flow velocity is close to zero if the fluctuation mode is modal or pressed. In the breathy phonation, however, there is a non-zero leakage component in the glottal flow during the entire duration of the glottal cycle. Hence, estimation of the vocal tract filter and lip radiation in different modes is a challenging task. But an accurate estimate of the filter can be obtained using the samples in the glottal closed phase, which are less affected by the glottal flow or the sub-glottal tube. The instant at which the glottis starts opening is called the glottal opening instant (GOI) and the instant at which the glottal flow derivative reaches its minimum is called the glottal closing instant (GCI). The GCIs and GOIs give the information about the closed phase, which helps to get an accurate estimation of the filter.

There are several algorithms in the literature for GIF. Since 1970s, several GIF methods have been proposed. Closed phase inverse filtering (CPIF) [8] estimates a parametric model of the spectral envelope, computed during the estimated closed phase duration. The challenge is to estimate the closed phase accurately. Normally the differentiated Electro Glottograph signal is used to find the closed phase. For example, Drugman *et al.* used the GCI and GOI information to locate the closed phase [9]. But the main disadvantage of the method is that, in high pitched voices, the closed phase duration is very small and this results in a poor estimate of the filter parameters. Iterative Adaptive

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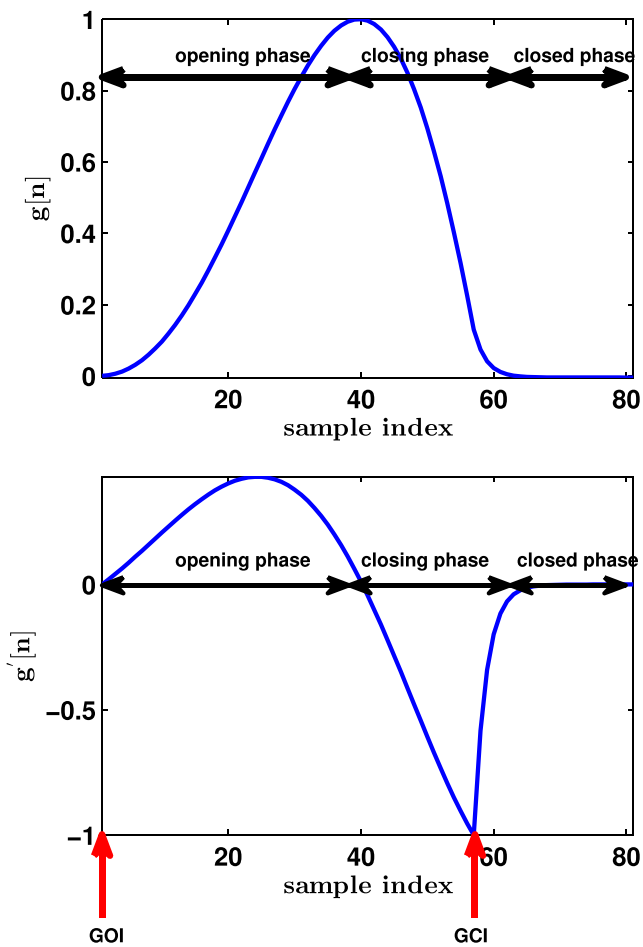

Fig. 1. Different stages of one cycle Glottal flow $g[n]$ (on top) and its derivative $g'[n]$ (below).

Inverse Filtering (IAIF) [10] utilizes an iterative scheme for estimating the glottal flow and the vocal tract spectral envelopes. The main advantage of IAIF is that the GCI/GOI location information is not required to estimate the GIF output. The method proposed by Moore *et al.* uses CPIF method and the refinement of GCIs using the peaks in linear prediction (LP) residual signal in an iterative manner to estimate the glottal flow. Hence, it is not very important to have an accurate GCI location estimate [11]. The original Zeros of the Z-Transform method [12], and the more recent, computationally more effective Complex Cepstrum Decomposition (CCD) method [13] separate the contributions of the glottal source and the vocal tract based on decomposing speech into minimum-phase and maximum-phase components within one cycle of the speech. The maximum phase component corresponds to the glottal flow. The method proposed by Auvinen *et al.* [14] uses a Markov chain Monte Carlo (MCMC) algorithm to estimate the RosenbergKlatt model parameters for GIF. Zhang *et al.* proposed a probabilistic model, where pitch, group delay, glottal vibration, vocal tract transfer function, and voiced/unvoiced amplitude are jointly modeled. The glottal flow is approximated by a third order all-pole filter and MCMC algorithm is used for inference [15], [16]. The method proposed by Ma *et al.* [17] modifies the conventional LP by introducing a temporal weighting on the LP residual and this method is referred to as the weighted linear prediction (WLP). Ma *et al.* [17] uses the short term energy of the signal for the temporal weighting. The WLP is shown to be less sensitive to the harmonic structure of the speech than conventional LP. The Quasi closed phase (QCP) method [7] uses the WLP to perform the inverse filtering using a special temporal window, called attenuate main excitation (AME), based the GCI locations. The GIF based on quadratic programming (QPR) [18] is motivated by QCP and uses both GCIs/GOIs to minimize the energy in the closed phase of the glottal flow. The lip radiation and the vocal tract filters are jointly estimated using quadratic programming. The methods QPR and QCP have been shown to be the state-of-the art methods for the GIFs. But one of the limitations of these methods is that they are sensitive to the GCI/GOI estimation errors and accurately detecting the closed phase is challenging. Hence, we hypothesize that estimating the closed phase directly from the speech is better than using the GCIs/GOIs.

In this paper we convert the WLP formulation into probabilistic WLP (PWLP), which, in turn, allows us to determine the probabilistic temporal weighting in a data-driven manner. We propose to use a temporal weight as convolution of a binary random variable and a fixed rectangular window. The binary random variable models the events similar to the GCIs/GOIs. For parameter estimation, we impose a Gaussian prior on the LP-filter coefficients and a Bernoulli prior on the binary random variable. We construct the posterior distribution using the prior and the likelihood of the PWLP. But the maximization of the posterior with respect to the binary random variable and the filter coefficients, in general, turns out to be intractable and, hence, we use Gibbs sampling method to sample from the posterior distribution and estimate the LP-filter coefficients. The GIF output is obtained by inverse filtering the speech using the estimated filter coefficients. The proposed method is called as probabilistic inverse filtering (PIF). For evaluation of the proposed method, we have experimented with three kinds of data 1) synthetic sustained vowel data using LF model [19] with different parameters and the formants of the five vowels (/ɑ/,/ɪ/,/æ/,/ʊ/,/ʌ/) synthesized using a linear source filter model with the fundamental frequencies varied from 80 Hz to 450 Hz and 2) three sustained vowels (/ɑ/,/ɪ/ and æ/) synthesized using a non-linear computational model of the human speech production [20] with the fundamental frequencies varied from 100 Hz to 450 Hz. 3) real speech data from five sustained vowels (/ɑ/,/ɪ/,/æ/,/ʊ/,/ʌ/) spoken by three male and three female speakers. The results show that the Normalized Amplitude Quotient (NAQ) obtained by the proposed method is better than that by QCP and IAIF by 0.035 and 0.104 respectively on the LF model based synthetic data. The NAQ for the proposed method is better than QCP by 0.12 on data obtained using the nonlinear computational model of the human speech production. The results based on real speech data show that the glottal flow estimated by the proposed method in the closed phase is flatter and has less formant ripple compared to QCP and QPR. Further, we demonstrate two main features of the proposed method: 1) the closed phase is automatically detected by the binary random variable at differ-

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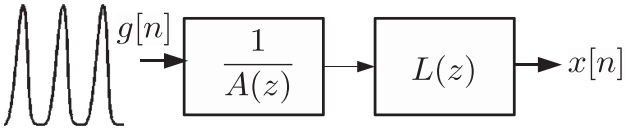

Fig. 2. Source filter model of speech.

ent ranges of fundamental frequencies and 2) the advantage of using Gaussian prior on the filter coefficients when the closed phase duration is small.

II. Proposed Approach

First we briefly review the WLP used in GIF, following which we give a probabilistic interpretation of the WLP. Then, we present the proposed probabilistic temporal weighting using binary random variable and a fixed window. We impose the priors on binary random variable and the filter coefficients and construct a posterior distribution to estimate the parameters.

A. Glottal Inverse Filtering Using Weighted Linear Prediction

The source-filter model of speech is shown in Fig. 2. The glottal excitation is passed through vocal-tract filter and lip-radiation filter to get the speech $x[n]$. The Z-transform of the $x[n]$ is given by

$$X(z) = \frac{G(z)L(z)}{A(z)}$$

where $G(z)$ is the Z-transform of the glottal signal $g[n]$, $\frac{1}{A(z)}$ is the transfer function of vocal tract filter and it is assumed to be all-pole, $A(z) = 1 - \sum_{k=1}^{K} a_k z^{-k}$, $L(z)$ is transfer function of the lip radiation. The derivative of the glottal flow is indicated by $g'[n]$. The transfer function of the lip radiation filter $L(z)$ is usually assumed to be a leaky differentiator given by

$$\hat{L}(z) = 1 - \zeta z^{-1},$$

where ζ is a constant within the range [0.96,1]. Hence, the lip radiation effect is removed from the speech using a pre-emphasis filter to get the signal $s[n]$.

$$S(z) \approx \frac{G(z)}{A(z)},$$

where $S(z)$ is the Z-transform of the signal $s[n]$. Thus, in GIF, it is required to obtain an accurate estimate of $A(z)$ from $s[n]$.

An accurate estimation of the $A(z)$ can be obtained by selecting a temporal weighting on $s[n]$ where the effect of $G(z)$ is minimal [7]. One way of achieving this is WLP, as done in QCP and QPR methods. WLP uses a prediction model similar to the conventional LP:

$$s[n] = \sum_{k=1}^{K} a_k s[n-k] + \eta[n], 0 \le n \le N_w - 1$$

where $s[n]$ is the speech frame of length N_w and $\eta[n]$ is the prediction error. When $n < 0$, the s values are from the previous frame. This can be compactly written as

$$\mathbf{s} = S\mathbf{a} + \eta, \tag{1}$$

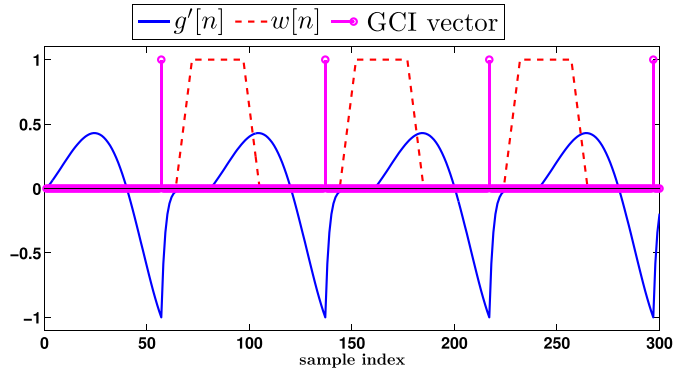

Fig. 3. Construction of window based on the GCIs, done in a manner similar to the QCP method.

where $s = [s[0], s[1], \ldots, s[N_w - 1]]^T$, $\mathbf{a} = [a_1, a_2, \ldots, a_K]^T$, $\eta = [\eta[0], \eta[1], \ldots, \eta[N_w - 1]]^T$ and

$$S = \begin{bmatrix} s[-1] & s[-2] & \ldots & s[-K] \\ s[0] & s[-1] & \ldots & s[-K+1] \\ \vdots & \vdots & \vdots & \vdots \\ s[N_w - 2] & s[N_w - 3] & \ldots & s[N_w - K + 1] \end{bmatrix}$$

where $[.]^T$ indicates the transpose operator. WLP differs from conventional LP in the sense that it imposes temporal weighting ($w[n]$) on the square of the residual:

$$E = \sum_{n=0}^{N_w - 1} \left(s[n] - \sum_{k=1}^{K} a_k s[n-k] \right)^2 w[n]. \tag{2}$$

This can be compactly written as

$$E = (\mathbf{s} - S\mathbf{a})^T W (\mathbf{s} - S\mathbf{a}) \tag{3}$$

where $W = diag(w)$ and $w = [w[0], w[1], \ldots, w[N_w - 1]]^T$. The method proposed by Ma *et al* [17] uses $w[n]$ based on the signal energy. In QCP method [7] the $w[n]$ is constructed based on the estimated GCIs such that the weight is lower around GCIs and, hence, the effect of the glottal pulse is reduced as shown in Fig. 3. The estimate of the $\mathbf{a}$ is obtained by minimizing the objective function given in eq. 3.

B. Probabilistic Weighted Linear Prediction (PWLP)

We assume that the distribution of η is $\mathcal{N}(\eta; 0, \sigma^2\Sigma)$. From eq. (1), the distribution of $\mathbf{s}$ can be written as:

$$p(\mathbf{s}|\{\mathbf{a}, \Sigma, \sigma^2\}) = \frac{1}{\sqrt{(2\pi)^d |\Sigma|}} e^{-\frac{1}{2} \frac{(\mathbf{s} - S\mathbf{a})^T \Sigma^{-1} (\mathbf{s} - S\mathbf{a})}{\sigma^2}}, \tag{4}$$

where $\sigma^2\Sigma$ indicates the covariance of the Gaussian noise η and $\mathcal{N}(\mathbf{z}; \mu, \Psi)$ indicates the multivariate Gaussian (for a d-dimensional vector $\mathbf{z}$) with mean μ and covariance Ψ given by

$$\mathcal{N}(\mathbf{z}; \mu, \Psi) = \frac{1}{\sqrt{(2\pi)^d |\Psi|}} e^{-\frac{1}{2}(\mathbf{z} - \mu)^T \Psi^{-1} (\mathbf{z} - \mu)}.$$

For $\Sigma = I$, the maximum likelihood (ML) estimate of $\mathbf{a}$ given $\mathbf{s}$ is same as the covariance LPC estimate [1]. For $\Sigma^{-1} = W$, the ML estimate of $\mathbf{a}$ turns out to be the WLP given in eq. (3).

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C. Probabilistic Model

We define the weighting function $w[n]$ as the function of an unknown binary random variable $\mathbf{b}$ and a fixed rectangular window of ones of length M as follows

$$w = A\mathbf{b},$$

where A is a $N_w \times N_w$ convolution matrix with first row being $[\underbrace{1,1,\ldots,1}_{M\text{-times}}, \underbrace{0,0,\ldots,0}_{(N_w-M)\text{-times}}]$, $\mathbf{b} = [b_0, b_1, \ldots, b_{N_w-1}]^T$ and b_k is a Bernoulli random variable. For PWLP (eq. (4)), the inverse covariance matrix is given by $\Sigma^{-1} = diag(w) = diag(A\mathbf{b}) = A\mathbf{b}e^T \odot I_n$, where e is a column vector of ones of length N_w, I_n indicates an identity matrix of size N_w and $\odot$ indicates the hadamard product. From eq. (4), the likelihood of the observed speech is given by

$$p(\mathbf{s}|\Theta) = \frac{1}{\sqrt{(2\pi)^d|\Sigma|}} e^{-\frac{(S\mathbf{a}-\mathbf{s})^T \Sigma^{-1}(S\mathbf{a}-\mathbf{s})}{2\sigma^2}} \tag{5}$$

$$= \frac{1}{\sqrt{(2\pi)^d\sigma^2|A\mathbf{b}e^T \odot I_n|^{-1}}} e^{-\frac{(S\mathbf{a}-\mathbf{s})^T\left[A\mathbf{b}e^T \odot I_n\right](S\mathbf{a}-\mathbf{s})}{2\sigma^2}} \tag{6}$$

where $d = N_w$, $\Theta = \{\mathbf{a}, \mathbf{b}, \sigma^2\}$. In general, the number of parameters $(N_w + K + 1)$ is more than that (N_w) of the observed speech samples $\mathbf{s}$ for parameter estimation. Hence, we impose a prior distribution on each of the model parameters. The details of the priors are described next.

1) Parameter Priors:

a) The b_k is assumed to be a Bernoulli random variable with parameter λ.

$$p(b_k) = Be(b_k; \lambda), \tag{7}$$

where $Be(z;\lambda)$ indicates the Bernoulli distribution with parameter λ defined as

$$Be(z;\lambda) = \begin{cases} p(z=1) = \lambda \\ p(z=0) = 1-\lambda \end{cases}$$

b) Filter coefficients prior: The prior distribution of filter coefficients $\mathbf{a}$ is assumed to be a zero mean Gaussian with covariance $\sigma_c^2 I$ and given by

$$p(\mathbf{a}) = \mathcal{N}(\mathbf{a}; \mathbf{0}, \sigma_c^2 I). \tag{8}$$

When the closed phase of the glottal cycle is short, the weighted prediction tends to become ill-conditioned [21] and results in a poor estimate of $\mathbf{a}$. This prior makes sure that the values of $\mathbf{a}$ do not go very large. This is similar to l_2-regularized linear prediction [21].

c) Noise variance prior: The prior distribution of noise variance is assumed to be the inverse Gamma distribution with parameters α and β.

$$p(\sigma^2) = \mathcal{IG}(\sigma^2; \alpha, \beta) \tag{9}$$

where $\mathcal{IG}(z;\alpha,\beta)$ indicates the inverse Gamma distribution with shape parameter α and scale parameter β given by

$$\mathcal{IG}(z;\alpha,\beta) = \frac{\beta^\alpha}{\Gamma(\alpha)} z^{-\alpha-1} e^{\frac{-\beta}{z}}, \quad z \geq 0$$

where $\Gamma(.)$ denotes the Gamma function [22].

We assume that the b_k, $\mathbf{a}$ and σ^2 are independent of each other and, hence, the joint distribution of the parameters is given by

$$p(\Theta) = \prod_{k=0}^{N_w-1} p(b_k)p(\mathbf{a})p(\sigma^2) \tag{10}$$

D. Parameter Estimation

Given the likelihood of speech in eq. (5) and the model parameters prior $p(\mathbf{\Theta})$ in eq. (10), the posterior distribution of the parameters is given by

$$p(\mathbf{\Theta}|s) \propto p(s|\mathbf{\Theta})p(\mathbf{\Theta}) = p(s|\mathbf{\Theta})p(\mathbf{a}) \prod_{k=0}^{N_w-1} p(b_k)p(\sigma^2) \tag{11}$$

The parameters $\mathbf{\Theta}$ are estimated by maximizing the posterior distribution in eq. (11) as follows:

$$\{\mathbf{b}^*, \mathbf{a}^*, \sigma^{2*}\} = \underset{\mathbf{a},\mathbf{b},\sigma^2}{\arg\max} \quad p(\mathbf{\Theta}|\mathbf{s}) \tag{12}$$

But directly maximizing the posterior distribution in eq. (11) is, in general, intractable. So we use Gibbs sampling [23] to generate the samples from the posterior distribution and then estimate the parameters. The Gibbs sampling is a method used to generate the samples from the joint distribution by iteratively sampling from the conditional distributions. Hence we derive the conditional distribution with respect to each model parameters given other parameters and the speech samples.

$$p(b_m = 1|\mathbf{s}, \mathbf{\Theta} \setminus \{b_m\}) = Be\left(\lambda \frac{\lambda_{1,m}}{\lambda\, \lambda_{1,m} + (1-\lambda)}\right) \tag{13}$$

$$p(\mathbf{a}|\mathbf{s}, \mathbf{\Theta} \setminus \{\mathbf{a}\}) = \mathcal{N}\left(\mathbf{a}; \mu, \left(\frac{S^T\Sigma^{-1}S}{\sigma^2} + \frac{I}{\sigma_c^2}\right)^{-1}\right) \tag{14}$$

$$p(\sigma^2|\mathbf{s}, \mathbf{\Theta} \setminus \{\sigma^2\}) = \mathcal{IG}(\sigma^2; \alpha + \frac{d}{2}, \frac{1}{2}\mathbf{d}^T\Sigma^{-1}\mathbf{d} + \beta) \tag{15}$$

where $\lambda_{1,m} = \frac{\sqrt{|\Sigma_k|_0^{-1}}}{\sqrt{|\Sigma_k|_1^{-1}}} e^{-z_k}$, $\mu = \frac{\left(\frac{S^T\Sigma^{-1}S}{\sigma^2} + \frac{I}{\sigma_c^2}\right)^{-1} S^T\Sigma^{-1}\mathbf{s}}{\sigma^2}$, $\mathbf{d} = \mathbf{s} - S\mathbf{a}$, $\mathbf{z} = \frac{e^T D_d D_d A}{2\sigma^2}$, $|\Sigma_k|_0$ is the determinant of the Σ when the $b_k = 0$ and z_k indicates the k-th element of the vector $\mathbf{z}$, D_d is diagonal matrix with the entries from $\mathbf{d}$. The derivations of the conditional distributions are given in Appendix. The steps of the Gibbs sampling to get the samples from the posterior distribution are shown in Algorithm 1. In Algorithm 1, $z[i]$ indicates the value of random variable z sampled in i-th iteration of Gibbs sampling. The number of samples generated by the Gibbs sampling is given by N_i and the number of burn-in[1] iteration is given by N_b.

[1]The burn-in period corresponds to the number of iterations of the sampler that is not used for estimating the parameters.

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Algorithm 1: Sampling Algorithm to Get Samples From eq. (11).

procedure SAMPLING ($\mathbf{s}, A, \mathbf{a_{init}}, \mathbf{b_{init}}, \sigma^2_{init}, N_i$)
  $\mathbf{a} \leftarrow \mathbf{a_{init}}$
  $\mathbf{b} \leftarrow \mathbf{b_{init}}$
  $\sigma^2 \leftarrow \sigma^2_{init}$
  $i \leftarrow 1$
top:
  for $k = 1 : N_w$ **do**
    sample $b_k[i]$ from $p(b_k|\mathbf{\Theta} \setminus \{b_k\}, \mathbf{s})$
  end for
  Construct $w = A\mathbf{b}$
  sample $\mathbf{a}[i]$ from $p(\mathbf{a}|\mathbf{\Theta} \setminus \{\mathbf{a}\}, \mathbf{s})$
  sample $\sigma^2[i]$ from $p(\sigma^2|\mathbf{\Theta} \setminus \{\sigma^2\}, \mathbf{s})$
  $i \leftarrow i + 1.$
  if $i > N_i$ **then return** $\mathbf{a}[i], 1 \leq i \leq N_i$
  end if
  goto *top*.
end procedure

Given the samples of the $\{\mathbf{a}[i], 1 \leq i \leq N_i\}$, the MAP estimate of $\mathbf{a}^*$ can be written as follows:

$$\mathbf{a}^* = \frac{1}{(N_i - N_b + 1)} \sum_{i=N_b}^{N_i} \mathbf{a}[i].$$

Given the samples $\{b_k[i], 1 \leq i \leq N_i, 0 \leq k \leq N_w - 1\}$, the MAP estimate of b_k can be written as follows:

$$b_k^* = \arg \max_{b_k \in \{0,1\}} q(b_k), \quad 0 \leq k \leq N_w - 1,$$

Where $q(b_k)$ is the relative multiplicity of the $b_k \in 0, 1$ in the samples $b_k[i]$.

E. Inverse Filtering

The final inverse filtered signal $\hat{g}[n]$ is obtained by inverse filtering the speech using the estimated $\mathbf{a}^*$.

$$\hat{g}[n] = s[n] - \sum_{k=1}^{K} a_k^* s[n-k]$$

III. Experiments and Results

A. Database

1) LF Model Test Data: The LF model is a parametric model for the glottal flow derivative waveform [24], where the t_e, t_p, t_a, E_e values define the shape of the waveform. The LF parameters also can be represented in an alternate dimensionless parameterization in terms of E_e, R_a, R_g, R_k, where the parameters can be interpolated. Four linearly spaced values are considered in the range corresponding to their span from breathy to creaky phonations [4]. This is done to sample the parameters in order to cover a wide range of possible excitations. The fundamental frequency was varied from 80 Hz to 450 Hz with a step of 10 Hz. This results in 9728 different excitation signals. The formants corresponding to the shapes of the vocal tract are obtained from [25], for simulating five different vowels (/ɑ/,/ɪ/,/æ/,/ʊ/,/ʌ/) with an all-pole filter with 6 poles (corresponding to 3 formants). The LF test contained a total of 48,640 test vowels each of duration 0.4 s at 8 kHz sampling rate. The test samples were divided into two categories based on their pitch: Low ($\leq$200 Hz) and High ($>$200 Hz) categories. This was done to investigate the performance of GIF at different frequency ranges.

2) Physical Model Test Data: The physical model test material is generated by physical modeling of the speech production mechanism. In natural human speech production, the independent source filter assumption may not hold good. Thus, physical model test examines the robustness of the proposed method when the test data does not come from a linear source-filter model. A thorough description of the physical model test data, used in this work, is available in [7], where the voice source component consisted of a kinematic representation of the medial surface of the vocal folds [26], [27] and the glottal flow waveform is affected by the interaction of the self-sustained oscillation of the vocal folds with subglottal and supraglottal pressures, similar to the one that occurs during real speech production [28], [29]. Three vowels (/ɑ/,/ɪ/ and æ/) are considered for evaluation in this work similar to that by Airaksinen *et al.* [7]. Eight pitch values ranging from 100 Hz to 450 Hz, with a step of 50 Hz, were used to synthesize these three vowels using physical model for the male, the female and the child. Vowel duration was 0.4 seconds and pitch was maintained constant during the utterance. Sampling frequency was set to 8 kHz.

3) Real Speech Test Data: We have used five sustained vowels (/ɑ/,/ɪ/,/æ/,/ʊ/,/ʌ/) spoken by three male and three female speakers from the CD available with Childers book [30]. The CD contains a parallel recoding of audio and the Electro-GlottoGraph (DEGG) signal. The sustained vowel recordings were high-pass filtered using a linear phase FIR filter with a cut-off frequency of 50 Hz for GIF analysis [10].

B. Evaluation

We use different evaluation metrics for synthetic data (LF model and the Physical model test data) and real speech data.

1) Synthetic Data: The objective evaluation of inverse filtering performance is done based on five different quantities, namely, NAQ, Quasi-Open Quotient (QOQ), H1H2, Harmonic Richness Factor (HRF), and mean squared error (MSE). These quantities describe estimated glottal flows from different perspectives. These are the same objective measures used in [7], [13], [18].

1) NAQ: The relative duration of the glottal closing phase is captured by the NAQ [31]. Its value is expected to indicate estimation errors that change the skewness of the flow pulse such as occurrence of severe formant ripple in the closed phase [7].
2) QOQ: QOQ is a measure for the relative open phase duration [32]. The estimation error in both opening and closing phase of the glottal cycle affects the QOQ.
3) H1H2 and HRF: H1H2 [33] and HRF [19] are computed using frequency domain representation. Estimation error

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anywhere in the glottal cycle affects the values of H1H2 and HRF. However, since both H1H2 and HRF measure the spectral decay of the glottal flow pulse waveform, a change in their values reflect mainly to the estimation error that affects the closing phase [7].

4) MSE: MSE is defined as a squared difference of the time-domain waveforms of the reference and the estimated glottal pulse derivative frames. It does not explicity depend on the glottal flow parameterization method, rather directly measures the dissimilarity between the original and estimated glottal flow.

For NAQ, QOQ and HRF, the difference between their reference values (NAQ_{ref}, QOQ_{ref}, HRF_{ref}) and the estimated ones (NAQ_{meas}, QOQ_{meas}, HRF_{meas}) are reported as average absolute relative error as follows:

$$ENAQ = E\left[\frac{|NAQ_{ref} - NAQ_{meas}|}{NAQ_{ref}}\right]$$

$$EQOQ = E\left[\frac{|QOQ_{ref} - QOQ_{meas}|}{QOQ_{ref}}\right]$$

$$EHRF = E\left[\frac{|HRF_{ref} - HRF_{meas}|}{HRF_{ref}}\right]$$

Similarly, the difference for H1H2 and MSE are reported as the average absolute error:

$$EH1H2 = E\left[|H1H2_{ref} - H1H2_{meas}|\right]$$
$$EMSE = E\left[(g_{ref} - g_{meas})^2\right],$$

where E indicates the sample average operator, g_{ref} is the grough truth glottal flow and g_{meas} is the estimated glottal flow.

2) Real Speech Data: We evaluate the performance of different methods using two evaluation metrics. As the first evaluation metric, the l_1-norm of the closed phase is computed similar to that in [18]. This metric is used to examine the behavior of the glottal flow in the closed phase. This is motivated by the fact that the closed phase is affected by the inaccurate estimation of the vocal tract filter. The closed phase is isolated by detecting the GCIs and GOIs from the Electroglottograph signal using SIGMA algorithm [34]. In order to compare the algorithms, the distribution of NAQ values, similar to [18], [35] is used as the second evaluation metric. This metric ensures that the performance is evaluated not necessarily only in terms of their closed phase behavior.

C. Baseline Schemes

For comparison, we use four baseline schemes, namely, QPR, QCP, CCD and IAIF. GCI estimate required for first three methods is obtained using the YAGA algorithm [36]. The MATLAB implementations of CCD and IAIF are obtained using the COVREP toolbox [37]. The QCP and QPR MATLAB implementations were provided by the respective authors.

D. Experimental Setup

A test speech signal is divided into frames of 30 ms duration ($N_w = 240$) with a shift of 10 ms. Prior to framing, the speech signal is pre-emphasized with $\zeta = 0.99$. We use a 10-th order filter ($K = 10$). For temporal weighting, the rectangular window length M is set to 3. A set of 100 random LF model based synthetic data is selected as the validation set that is not used for evaluation. The hyper parameters σ_c, α and β are optimized on the validation set to minimize the $ENAQ$ using a grid search with following values for different parameters: $\log(\sigma_c)$ from -8 to 1 with a step of 1, α from 0 to $5N_w$ with a step of N_w and β from 0 to 3 with a step of 0.5. The optimum value of σ_c, α and β is found to be 0.01, $3N_w$ and 1 respectively. It has been empirically found that the $N_i = 60$ is sufficient for the convergence of Algorithm 1 and $N_b = 20$ is sufficient to get the samples from the distribution and λ is set to 0.03.

E. Analysis of Estimated Parameters

We demonstrate two main features of the PIF using representative examples in Fig. 4 and Fig. 5. The first feature is that the rectangular window location for temporal weighting is not set to a fixed value, rather it is learnt in a data-driven manner and adapts itself to the closed phase duration in each glottal cycle. Left column of Fig. 4 shows the original glottal derivative signal $g'[n]$ (in blue line), the estimated GIF output $\hat{g}'[n]$ (in red line), the temporal weighting function $w[n]$ found by PIF in the last iteration of Algorithm 1 (in black line), and the corresponding indicators where $b_k^* = 1$ (in magenta dots) in three analysis frames. The right column shows the speech spectrum, the frequency response of the original filter and the frequency response of the filter estimated by the PIF in those respective frames. Three rows in Fig. 4 correspond to three excitations which have different closed phase and fundamental frequency. The temporal weights estimated by PIF match closely with the closed phase duration and, hence, the estimated GIF signal is close to the original one. This is also reflected in the estimated filter frequency response (in red line Fig. 4), which closely matches with original one (in black). It is clear from the first row of Fig. 4 that, non-zero temporal weights are achieved in the closed phase of all glottal cycle including the very first one cycle resulting in a reliable estimate of the filter. However, in the baseline methods, unlike that in the proposed one, the temporal weight is determined based on the GCIs, which would miss the samples of the closed phase in the very first glottal cycle as the GCI for that cycle lies in the frame before the analysis frame shown in Fig. 4.

The second feature of the proposed PIF shows the importance of the Gaussian prior on the filter coefficients. Fig. 5 shows the original speech signal, the recovered GIF output for $\sigma_c = \{0.004,\ 0.01,\ 0.1,\ 100\}$. It is clear from the figure that as $\sigma_c \to 0$ the inverse filtered output becomes more like speech, because filters with zero coefficients are encouraged more than others. As the $\sigma_c \to \infty$ the inverse filtered output becomes more impulse like in nature, i.e., the method becomes similar to the LP method. The GIF output for $\sigma_c = 0.1$ matches more closely with the

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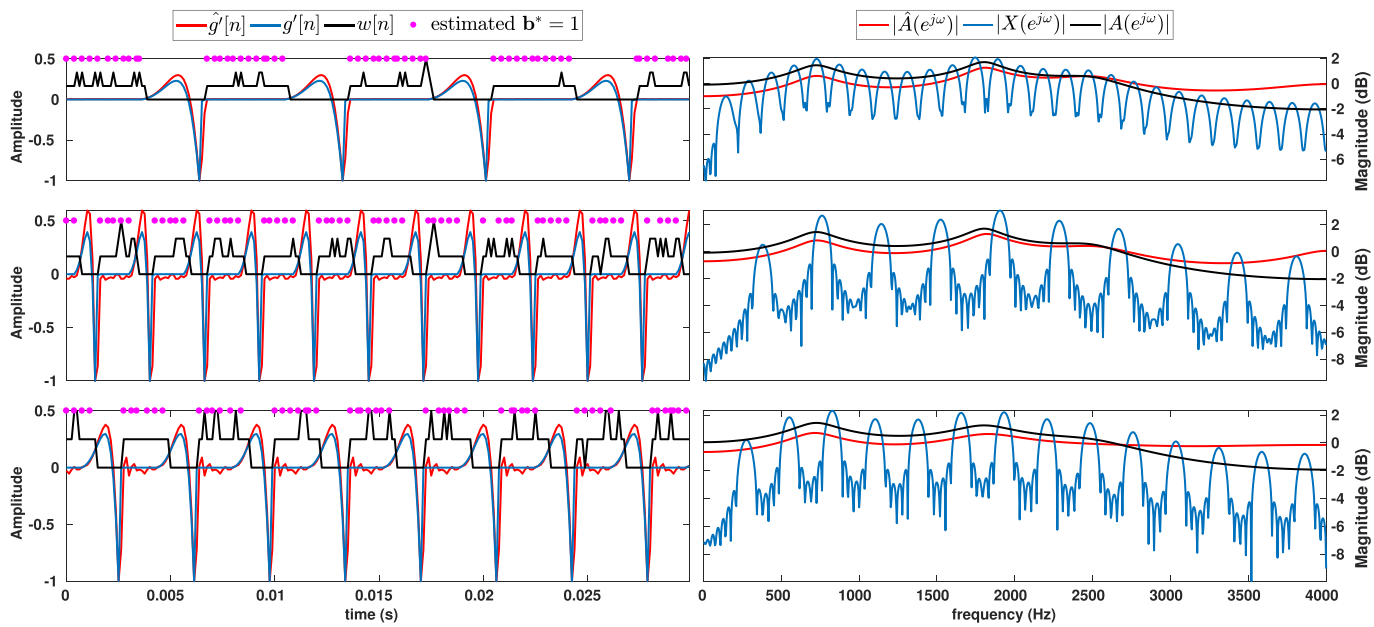

Fig. 4. Illustration of the automatic closed phase selection by the proposed PIF. Three excitations (corresponding to three rows) have been considered with fundamental frequencies 145, 375, 275 Hz respectively. $|\hat{A}(e^{j\omega})|$ is magnitude response of the estimated filter $\mathbf{a}^*$.

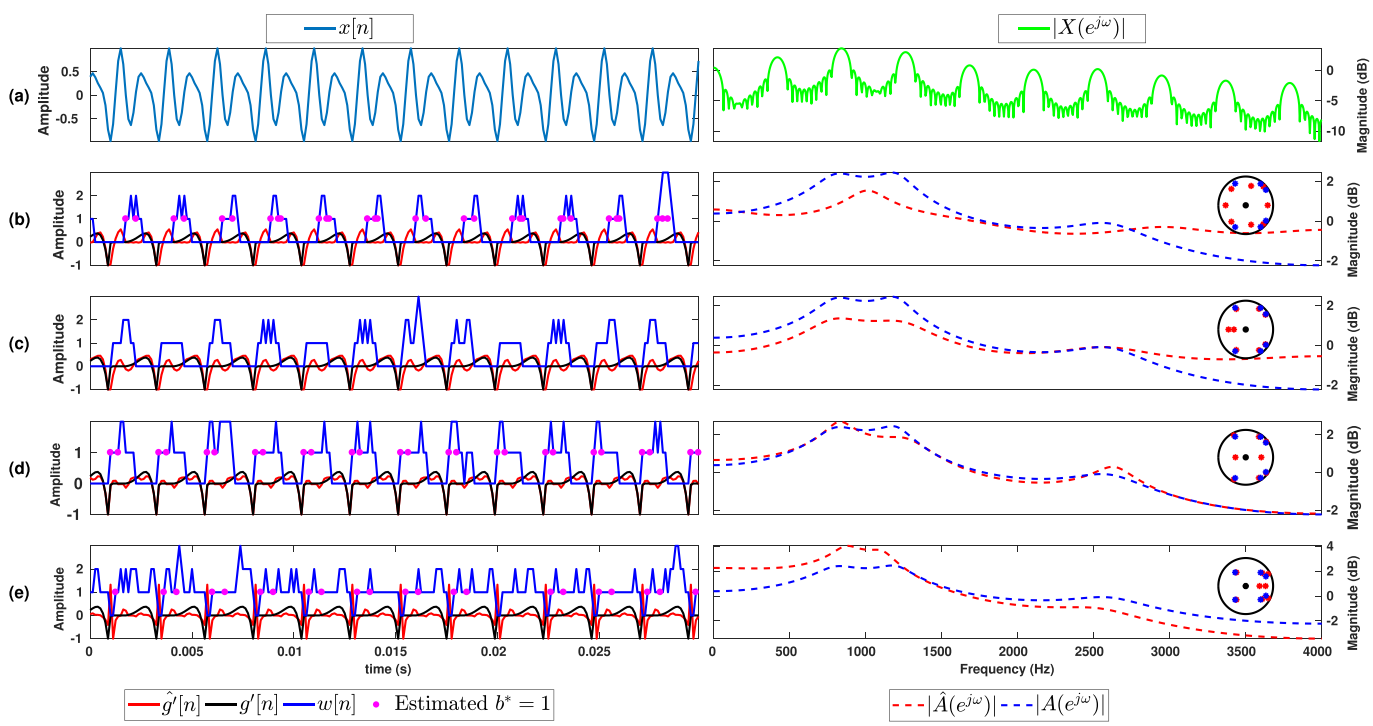

Fig. 5. Illustration of the effect of Gaussian prior on the Inverse filtering. (a) shows the speech signal and the corresponding magnitude spectrum. (b)–(e) show the ground truth glottal flow, the magnitude response ground truth filter, the estimated glottal flow and the magnitude response estimated filter for different value of σ_c. (b): $\sigma_c = .004$, (c): $\sigma_c = 0.01$, (d): $\sigma_c = 0.1$, (e): $\sigma_c = 100$. The poles of the original and estimated filters are also shown inside the unit circle in respective cases.

reference one compared to using other values of σ_c. This is clear from Fig. 5(d) where the temporal weights closely match with the closed phase and the estimated filter frequency response also closely matches with the original filter frequency response. It is clear from the last row of the figure that, temporal weights are non-zero for most of the samples in the frame and the estimated filter poles are closer to theunit circle compared to other smaller values of σ_c. On the other hand, as the $\sigma_c \to 0$, the poles near the zero are encouraged as clear from the polo-zero plots in Fig. 5(b).

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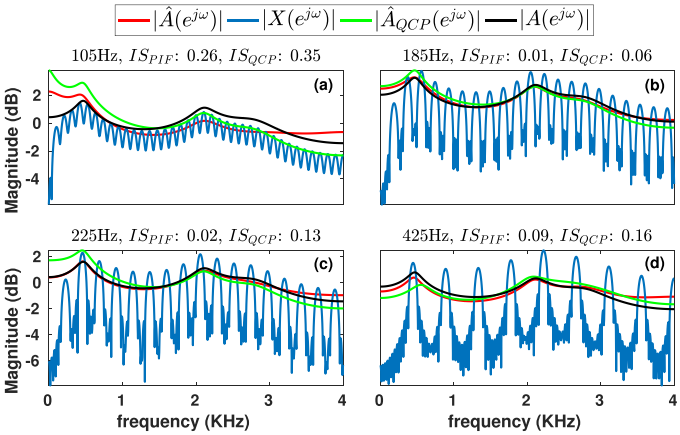

Fig. 6. Comparison of estimated filter magnitude response for vowel $/I/$ and different pitch values using QCP and PIF. $|\hat{A}(e^{j\omega})|$ is magnitude response of the estimated filter $\mathbf{a}^*$, $|\hat{A}_{QCP}(e^{j\omega})|$ is magnitude response of the estimated filter using QCP. Title of each figure show the pitch value, IS_{PIF} indicates the Itakura-Saito distance between $|A(e^{j\omega})|$ and $|\hat{A}(e^{j\omega})|$ and IS_{QCP} indicates the Itakura-Saito distance between $|A(e^{j\omega})|$ and $|\hat{A}_{QCP}(e^{j\omega})|$.

In Fig. 6, we illustrate the estimated filters obtainedusing PIF and QCP method. Four subplots show four different examples with the vowel$/I/$ and the following pitch values (a) 105 Hz, (b) 185 Hz, (c) 225 Hz and (d) 425 Hz. To quantify the differences in estimated filter magnitude response, we compute the Itakura-Saito distance [38] between the original and estimated filter magnitude responses. It is clear from the figure that the Itakura-Saito distance for PIF is lower than that for the QCP. It indicates that the PIF is able to estimate the spectrum more accurately. In Fig. 6 (a–c), gain of the first formant in the estimated filter using the QCP method is more than that using the PIF method. This could be because of a few numbers of samples to estimate the filter, but in PIF, the regularization helps in estimating the filter more accurately.

F. Results

Fig. 7 shows the comparison of the PIF method with the baseline schemes for LF model based synthetic data and the physical model based synthetic data using different evaluation metrics mentioned in the Section III-B1. For each evaluation metric and GIF method, the test data is divided into three categories based on the fundamental frequency range, namely, 1) $f0 \leq 200$, 2) $f0 > 200$ 3) all.This is done to compare the relative performance of every method in different ranges of fundamental frequency.

From the first row of Fig. 7. it can be observed that the PIF is better than the baseline schemes in all the evaluation metrics for LF model based synthetic data. The improvement is higher for $f0 > 200$ than that for $f0 \leq 200$. The high fundamentalfrequency results in a lower closed phase duration. Higher performance gain in high f 0 using the PIF method indicates that the proposed method is able to estimate the filter more reliably than the baseline schemes in high f 0 condition. The average ENAQ of PIF is better than QCP and IAIF by 0.035and 0.104 respectively. The NAQ improvement is higher in high fundamental frequency. We observe that this improvement is mainly because of the accurate estimate of the filter, which results in less formant ripples in the GIF.The EQOQ metric for PIF isbetter than the QCP and CCD by 0.36 (absolute) and 0.56 (absolute) respectively. At high fundamental frequency the PIF performs better than the QCP by 0.008 (absolute). The EH1H2 for PIFisbetter than the QCP by 0.69 (absolute). The EHRF for PIF is better than CCD and QCP by 2.92 (absolute) and 5.4 (absolute)respectively. We observe that for CCD, the estimated GIF output is close to impulses which cause the HRF to be very high and result in poor ENAQ and EQOQ. At lower fundamental frequency, EHRF for PIF is 14.83 (absolute) lower as compared to the QCP.The MSE for PIF is lower than that of the baseline schemes. Overall, it can be noticed that the PIF is not significantly affected by different $f0$ ranges.

Fig. 7 (second row) shows the comparison of the PIF method with the baseline schemes for physical model based synthetic test data. Most of the metrics for this data is higher compared to theLF model based synthetic data, because all methods assume linear source filter model. We see that the PIF is more affected, especially in the high fundamental frequency region. This could be mainly because the value of σ_c is learnt from the LF model based synthetic data. The ENAQ for PIF is lower than QCP by 0.124 (absolute), indicating that the PIF is able to reduce the formant ripples in the closed phase compared to baseline schemes. But the EQOQ for PIF is higher than that for QCP by 0.68 (absolute), which indicates that the open/closed phase of the GIF output is poor in PIF.The EHRF and EH1H2 for PIF are higher than QCP because we observe that the estimated closing phase for the PIF is poor . The EMSE for PIF is lower than the QCP by 0.05 (absolute). This could be because the ripple in closed phase estimated by the QCP is higher compared to that by the PIF.

Fig. 8 shows a comparison of the PIF method with the baseline schemes in terms of theirperformances on real speech data. Fig. 8(a) shows a comparison of the average l_1- norm of the closed phase. It is clear from the figure that the PIF method results in the lowest average l_1-norm of the closed phase although such a metric was not directly used in the objective function of the PIF method. Lower l_1-norm by PIF method over other baseline schemes, indicates that the estimated filters are more accurate using the proposed PIF method. We observe that the average l_1- norm for QPR is high mainly because of the errors in GCI/GOI. Fig. 8(b) shows the comparison of the NAQ distribution for different methods. It can be observed from the figure that the IAIF, QCP and PIF follow similar distribution. But the NAQ values obtained with the QPR method differ to a large extent from the other GIF methods. In the case of QPR, some high NAQ values are observed primarily due to GCI/GOI errors. Fig. 9 shows four examples of the glottal flow estimated from PIF and QPR on four vowels /ɪ/,/ u /,/ʌ/,/ɑ/. QPR is used for comparison in Fig. 9 because it has the lowest average l_1-norm among the baseline schemes. It is clear from Fig. 9(a) and (b) that the PIF has flatterclosed phase and lesser formant ripple compared to the QPR method. It is clear from Fig. 9(d) that the QPR has flatter closed phase compared to the PIF. Thiscould be because the PIF uses a constant lip radiation compensation.

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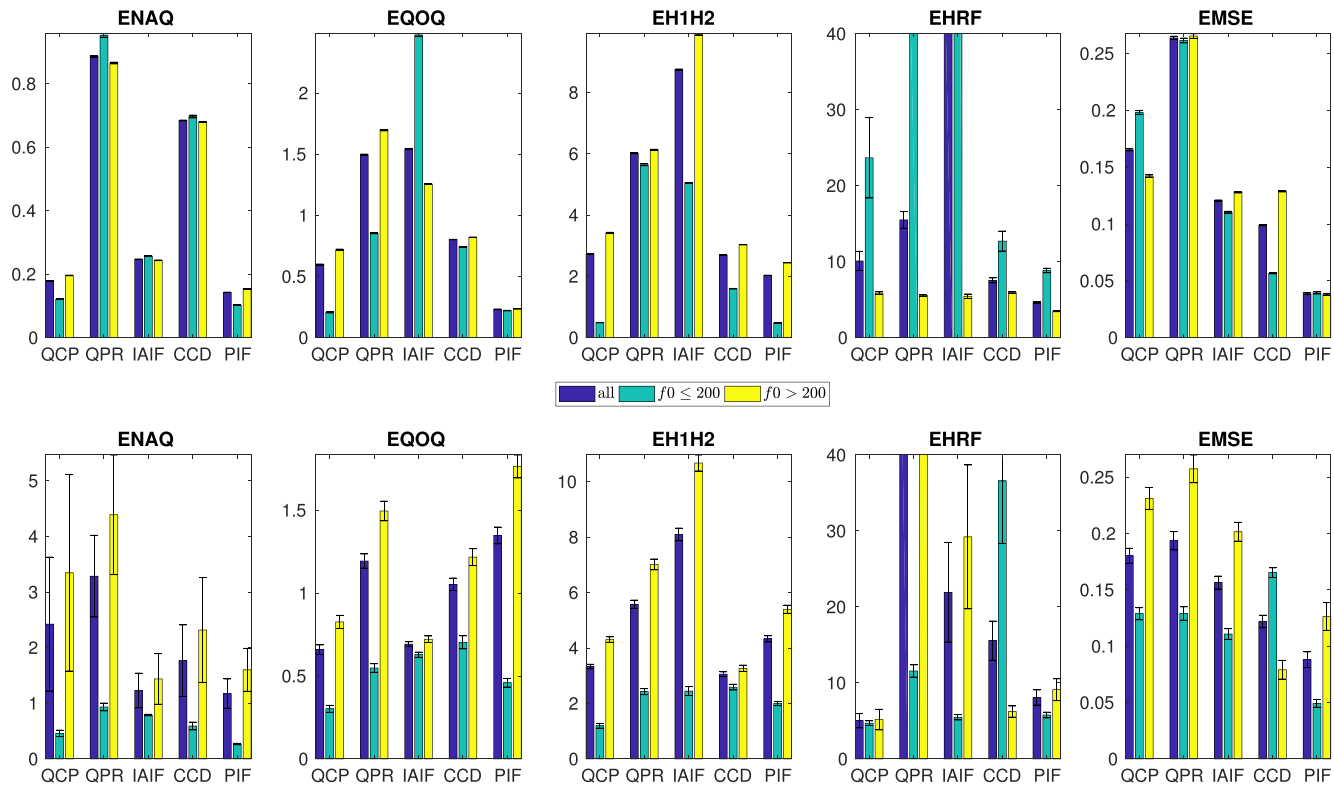

Fig. 7. Comparison of the baseline methods and the proposed PIF method using five different evaluation metrics for the LF model based synthetic data (top row) and the physical model based synthetic data (bottom row). Error bar indicates the 95% confidence interval.

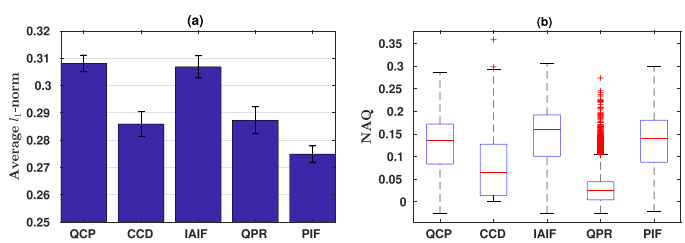

Fig. 8. Results on real speech test data: (a): Comparison of average (with errorbar indicating 95% confidence interval) of l_1 - norm of the closed phase for baselines and the proposed PIF method. (b): Boxplot of the NAQ distribution for baselines and the proposed PIF method.

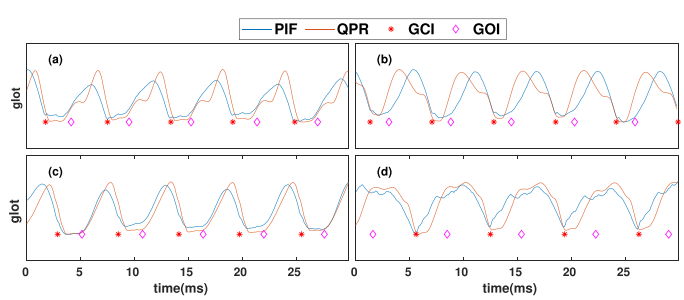

Fig. 9. Comparison of four representative examples of the estimated glottal flow (glot) from the real speech using PIF and QPR. Fig. (a)-(d) represent glottal flow estimated from thevowels /ɪ/,/ u/,/ʌ/,/ɑ/ respectively.

IV. Conclusion

We proposed a glottal inverse filtering method based on PWLP and the probabilistic temporal weighting function, where the weight is defined in terms of binary vector and a fixed window. We constructed a posterior distribution based on the PWLP likelihood and imposing a Gaussian prior on the filter coefficients. The parameters are estimated using the Gibbs sampling. We showed that the proposed PIF performs better than the baseline for the LF model based synthetic data. In case of physical model based synthetic data, PIF performs better only in NAQ and MSE compared to the baseline methods. For real speech data, the PIF has a lower average l_1-norm in the closed phase compared to other methods. We showed that the PIF is free of GCI detection (hence closed phase). We also showed the advantage of using the Gaussian prior on the filter coefficients. The proposed method can be further improved on physical model based synthetic data by imposing the hyper prior on the σ_c, λ and learning them from the data. The temporal weights learnt from the data can be used detect the GCIs and GOIs. Noise robustness of the PIF method requires a thorough experimental investigation. Investigation is also required to examine the hyper parameter dependency on the different types of data. These are parts of our future works.

Appendix

We will use the following identity in the derivations below and we use $d = N_w$.

$$\int_{\mathbf{x}} e^{-\frac{1}{2}\mathbf{x}^T A\mathbf{x}+\mathbf{b}^T\mathbf{x}} \mathrm{d}^n\mathbf{x} = \sqrt{\frac{(2\pi)^n}{det(A)}} e^{\frac{1}{2}\mathbf{b}^T A^{-1}\mathbf{b}} \tag{16}$$

$\mathrm{d}^n\mathbf{x}$ is the n-dimensional volume differential.

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Consider $\Sigma^{-1} = diag(A\mathbf{b}) = A\mathbf{b}e^T \odot I_n$ where e is the vector of ones and I_n indicates the identity matrix of size n. $\mathbf{d}^T \left[A\mathbf{b}e^T \odot I_n\right] \mathbf{d} = tr\{D_d A\mathbf{b}e^T D_d\} = tr\{e^T D_d D_d A\mathbf{b}\} = e^T D_d D_d A\mathbf{b}$, where D_d is a diagonal matrix with the entries from $\mathbf{d}$. The eq. 5 can be written as

$$p(\mathbf{s}|\Theta) = \frac{1}{\sqrt{c\sigma^2 |A\mathbf{b}e^T \odot I_n|^{-1}}} e^{-\frac{e^T D_d D_d A\mathbf{b}}{2\sigma^2}} \tag{17}$$

where $c = (2\pi)^d$, $\mathbf{d} = \mathbf{s} - S\mathbf{a}$ and $d = N_w$.

A. Sampling Distributions

$$p(b_k = 1|\mathbf{s}, \mathbf{\Theta}\backslash b_k) = \frac{p(\mathbf{\Theta}, \mathbf{s})}{\sum_{b_k \in \{0,1\}} p(\mathbf{\Theta}, \mathbf{s})}$$

$$= \frac{p(\mathbf{s}|\mathbf{\Theta})p(\mathbf{\Theta})}{\sum_{b_k \in \{0,1\}} p(\mathbf{s}|\mathbf{\Theta}\backslash b_k, b_k)p(\mathbf{\Theta})}$$

$$= \frac{p(\mathbf{s}|\mathbf{\Theta}) \prod_{l=0}^{N_w-1} p(b_l)p(\mathbf{a})p(\sigma^2)}{\sum_{b_k \in \{0,1\}} p(\mathbf{s}|\mathbf{\Theta}\backslash b_k, b_k) \prod_{l=0}^{N_w-1} p(b_l)p(\mathbf{a})p(\sigma^2)}$$

Canceling the terms indepedent of k and from eq. 17 and eq. 7

$$= \frac{p(\mathbf{s}|\mathbf{\Theta})p(b_k)}{\sum_{b_k \in \{0,1\}} p(\mathbf{s}|\mathbf{\Theta}\backslash b_k, b_k)p(b_k)}$$

$$= \frac{\frac{\lambda}{\sqrt{c\sigma^2 |A\mathbf{b}e^T \odot I_n|^{-1}}} e^{-\frac{e^T D_d D_d A\mathbf{b}}{2\sigma^2}}}{\sum_{b_k \in \{0,1\}} \frac{1}{\sqrt{c\sigma^2 |A\mathbf{b}e^T \odot I_n|^{-1}}} e^{-\frac{e^T D_d D_d A\mathbf{b}}{2\sigma^2}} p(b_k)}$$

$$= \frac{\frac{\lambda\sqrt{|\Sigma_k|_0^{-1}}}{\sqrt{|\Sigma_k|_1^{-1}}} e^{-z_k}}{\frac{\lambda\sqrt{|\Sigma_k|_0^{-1}}}{\sqrt{|\Sigma_k|_1^{-1}}} e^{-z_k} + 1 - \lambda}$$

$$= Be\left(\frac{\lambda\lambda_{1,m}}{\lambda\,\lambda_{1,m} + (1-\lambda)}\right)$$

where $\mathbf{z} = \frac{e^T D_d D_d A}{2\sigma^2}$, $|\Sigma_k|_0$ is the determinant of the Σ when the $b_k = 0$ and z_k indicates the k-th element of the vector $\mathbf{z}$ and $\lambda_{1,m} = \frac{\sqrt{|\Sigma_k|_0^{-1}}}{\sqrt{|\Sigma_k|_1^{-1}}} e^{-z_k}$.

$$p(\mathbf{a}|\mathbf{\Theta}\backslash\{\mathbf{a}\}, \mathbf{s}) = \frac{p(\mathbf{s}, \mathbf{\Theta})}{\int_{\mathbf{a}} p(\mathbf{s}, \mathbf{\Theta})\mathrm{d}\mathbf{a}} = \frac{p(\mathbf{s}|\mathbf{\Theta})p(\mathbf{\Theta})}{\int_{\mathbf{a}} p(\mathbf{s}|\mathbf{\Theta})p(\mathbf{\Theta})\mathrm{d}\mathbf{a}}$$

$$= \frac{p(\mathbf{s}|\mathbf{\Theta}) \prod_k p(b_k)p(\sigma^2)p(\mathbf{a})}{\int_{\mathbf{a}} p(\mathbf{s}|\mathbf{\Theta}) \prod_k p(b_k)p(\sigma^2)p(\mathbf{a})\mathrm{d}\mathbf{a}}$$

cancelling the terms that are independent of $\mathbf{a}$

$$= \frac{\frac{\sigma^{\frac{d}{2}}}{(2\pi)^{\frac{d+K}{2}} |\Sigma|^{\frac{1}{2}} \sigma_c^{\frac{K}{2}}} e^{-\frac{(S\mathbf{a}-\mathbf{s})^T \Sigma^{-1} (S\mathbf{a}-\mathbf{s})}{2\sigma^2} - \frac{\mathbf{a}^T\mathbf{a}}{2\sigma_c^2}}}{\frac{\sigma^{\frac{d}{2}}}{(2\pi)^{\frac{d+K}{2}} |\Sigma|^{\frac{1}{2}} \sigma_c^{\frac{K}{2}}} \int_{\mathbf{a}} e^{-\frac{(S\mathbf{a}-\mathbf{s})^T \Sigma^{-1} (S\mathbf{a}-\mathbf{s})}{2\sigma^2} - \frac{\mathbf{a}^T\mathbf{a}}{2\sigma_c^2}} \mathrm{d}\mathbf{a}} \tag{18}$$

Consider $-\frac{(S\mathbf{a}-\mathbf{s})^T \Sigma^{-1} (S\mathbf{a}-\mathbf{s})}{2\sigma^2} - \frac{\mathbf{a}^T\mathbf{a}}{2\sigma_c^2} = -\frac{1}{2}\mathbf{a}^T \left(\frac{S^T\Sigma^{-1}S}{\sigma^2} + \frac{I}{\sigma_c^2}\right)\mathbf{a} + \frac{\mathbf{a}^T S^T \Sigma^{-1}\mathbf{s}}{\sigma^2} + C_2$, where C_2 is a constant independent of $\mathbf{a}$ and common in both numerator and denominator.Using eq. (16),

$$\int_{\mathbf{a}} e^{-\frac{1}{2}\mathbf{a}^T \left(\frac{S^T\Sigma^{-1}S}{\sigma^2} + \frac{I}{\sigma_c^2}\right)\mathbf{a} + \frac{\mathbf{a}^T S^T \Sigma^{-1}\mathbf{s}}{\sigma^2}} \mathrm{d}\mathbf{a}$$

$$= \sqrt{(2\pi)^K |P|^{-1}} e^{\frac{\mathbf{s}^T \Sigma^{-1} S P^{-1} S^T \Sigma^{-1}\mathbf{s}}{2\sigma^4}}$$

where $P = \left(\frac{S^T\Sigma^{-1}S}{\sigma^2} + \frac{I}{\sigma_c^2}\right)$. Substitute in eq. (18)

$$p(\mathbf{a}|\mathbf{\Theta}\backslash\{\mathbf{a}\}, \mathbf{s})$$

$$= \frac{1}{\sqrt{(2\pi)^K |P|^{-1}}} e^{-\frac{1}{2}\left[\mathbf{a}^T P\mathbf{a} - \frac{2\mathbf{a}^T S^T \Sigma^{-1}\mathbf{s}}{\sigma^2} + \frac{\mathbf{s}^T \Sigma^{-1} S P^{-1} S^T \Sigma^{-1}\mathbf{s}}{\sigma^4}\right]}$$

$$= \mathcal{N}\left(\mathbf{a}; \mu, \left(\frac{S^T\Sigma^{-1}S}{\sigma^2} + \frac{I}{\sigma_c^2}\right)^{-1}\right) \tag{19}$$

where $\mu = \frac{\left(\frac{S^T\Sigma^{-1}S}{\sigma^2} + \frac{I}{\sigma_c^2}\right)^{-1} S^T\Sigma^{-1}\mathbf{s}}{\sigma^2}$

$$p(\sigma^2|\mathbf{\Theta}\backslash\{\sigma^2\}, \mathbf{s}) = \frac{p(\mathbf{s}, \mathbf{\Theta})}{\int_{\sigma^2} p(\mathbf{s}, \mathbf{\Theta})\mathrm{d}\sigma^2} = \frac{p(\mathbf{s}|\mathbf{\Theta})p(\mathbf{\Theta})}{\int_{\sigma^2} p(\mathbf{s}|\mathbf{\Theta})p(\mathbf{\Theta})\mathrm{d}\sigma^2}$$

$$= \frac{p(\mathbf{s}|\mathbf{\Theta}) \prod_k p(b_k)p(\sigma^2)p(c)}{\int_{\sigma^2} p(\mathbf{s}|\mathbf{\Theta}) \prod_k p(b_k)p(\sigma^2)p(c)\mathrm{d}\sigma^2}$$

cancelling the terms that are independent of the σ^2

$$= \frac{p(s|\mathbf{\Theta})p(\sigma^2)}{\int_{\sigma^2} p(s|\mathbf{\Theta})p(\sigma^2)\mathrm{d}\sigma^2}$$

$$= \frac{\frac{\beta^\alpha \left(\sigma^2\right)^{-\alpha-1-\frac{d}{2}}}{\Gamma(\alpha)(2\pi)^{\frac{d}{2}}} e^{-\frac{\mathbf{d}^T\Sigma^{-1}\mathbf{d}}{2\sigma^2} - \frac{\beta}{\sigma^2}}}{\frac{\beta^\alpha}{\Gamma(\alpha)(2\pi)^{\frac{d}{2}}} \int_{\sigma^2} \left(\sigma^2\right)^{-\left(\alpha+\frac{d}{2}\right)-1} e^{-\frac{\frac{1}{2}(\mathbf{d}^T\Sigma^{-1}\mathbf{d})+\beta}{\sigma^2}} \mathrm{d}\sigma^2} \tag{20}$$

From the definition of the $\mathcal{IG}$ distribution, the denominator can be expressed as

$$\frac{\beta^\alpha}{\Gamma(\alpha)(2\pi)^{\frac{d}{2}}} \int_{\sigma^2} \left(\sigma^2\right)^{-\left(\alpha+\frac{d}{2}\right)-1} e^{-\frac{\frac{1}{2}(\mathbf{d}^T\Sigma^{-1}\mathbf{d})+\beta}{\sigma^2}} \mathrm{d}\sigma^2$$

$$= \frac{\beta^\alpha}{\Gamma(\alpha)(2\pi)^{\frac{d}{2}}} \frac{\Gamma(\alpha+\frac{d}{2})}{\left\{\frac{1}{2}(\mathbf{d}^T\Sigma^{-1}\mathbf{d}) + \beta\right\}^{\alpha+\frac{d}{2}}}.$$

Substituting in eq. (20),

$$p(\sigma^2|\mathbf{\Theta}\backslash\{\sigma^2\}, \mathbf{s}) = \frac{\frac{\beta^\alpha \left(\sigma^2\right)^{-\alpha-1-\frac{d}{2}}}{\Gamma(\alpha)(2\pi)^{\frac{d}{2}}} e^{-\frac{\mathbf{d}^T\Sigma^{-1}\mathbf{d}}{2\sigma^2} - \frac{\beta}{\sigma^2}}}{\frac{\beta^\alpha}{\Gamma(\alpha)(2\pi)^{\frac{d}{2}}} \frac{\Gamma(\alpha+\frac{d}{2})}{\left\{\frac{1}{2}(\mathbf{d}^T\Sigma^{-1}\mathbf{d})+\beta\right\}^{\alpha+\frac{d}{2}}}}$$

$$= \frac{\left\{\frac{1}{2}(\mathbf{d}^T\Sigma^{-1}\mathbf{d}) + \beta\right\}^{\alpha+\frac{d}{2}} \left(\sigma^2\right)^{-\left(\alpha+\frac{d}{2}\right)-1} e^{-\frac{\frac{1}{2}\mathbf{d}^T\Sigma^{-1}\mathbf{d}+\beta}{2\sigma^2}}}{\Gamma(\alpha+\frac{d}{2})}$$

$$= \mathcal{IG}\left(\sigma^2; \alpha + \frac{d}{2}, \frac{1}{2}\mathbf{d}^T\Sigma^{-1}\mathbf{d} + \beta\right) \tag{21}$$

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